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Low-Cost X-band Doppler Radar for Nadir Velocity Detection

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***Abstract*— The low-cost, low power, small X-band (10.5 GHz) Doppler radar module HB100, was designed and is used commercially for movement detection. As with most radars, radial displacements of targets provide positive or negative Doppler shifts – depending whether they are approaching or receding in the radial direction. However, in many applications the movement tangential to the radial direction is of interest as well. The radar return from a short steel pipe (outer diameter 170 mm) rolling at a constant velocity on a horizontal plane was used to investigate the effect on the radar output as the pipe rolled through the E-plane and H-plane radar beams. Given the finite beamwidths (40 degrees and 80 degrees, respectively), the voltage output signal versus time is symmetrical with respect to the Nadir in both the approaching and receding part of the trajectory of the pipe for uniform displacement perpendicular to the radar beam. With a null recording immediately beneath the radar in both planes, the velocity of the pipe is monotonically related to the time between the two major peaks and troughs. This allows the velocity determination of objects moving through the radar beam tangentially.**

Keywords*— *Doppler radar, E-plane, H-plane, nadir detection, low cost, low power, movement velocity

I. Introduction

There are many applications important for the detection of movement through a radar beam. Such applications include motor vehicles, light switching, and water flow. The basic theory relating to Doppler radar is the detection of radial velocity which imparts a positive or negative frequency shift on the received signal. Commonly small, unobtrusive radars would be placed conformally with a mounting structure. As the radar has a finite beamwidth, a single moving scatterer should create a Doppler shift before and after the main beam direction. To resolve this ambiguity, most Doppler radars are oriented at an acute angle to the expected incoming direction of the target [1-4]. This paper reports the incoming and outgoing movement of a rolling steel pipe through both planes of the bistatic antenna array of an HB100 commercial Doppler radar at 10.525 GHz.

II. The HB100 Doppler Radar Module

The earliest published record describing the HB100 radar sensor was published in 2014 [5], with the description as a "miniature microwave sensor X-band bi-static Doppler transceiver module" with a range of 20 m with half-power beam widths of 80 degrees in azimuth (H-plane) and 40 degrees in elevation (E-plane). The module requires 4.75-5.25 V DC with a CW output signal at 10.525 GHz. The antenna configuration is a rectangular arrangement of four patches with linear polarization (Fig 1). The output signal from the module was amplified, sampled at 10,000 samples per second and logged by computer via a USB port. The radar module was held in position in a hole on a wooden plank and connected via a 2 m USB cable.

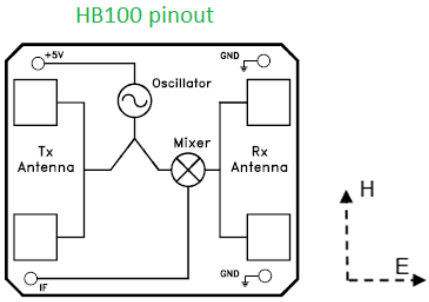


Fig. 1. Electrical connections and patch antenna array of the HB100 Doppler radar module showing the E and H planes of linear polarisation. The circuit board is 45 × 36 mm^2 on FR4.

III. Methods and Results

A round, steel pipe (outer diameter 170 mm) was released from a stationary position at the end of a rigid inclined ramp onto a flat rigid surface at various elevation heights (see Fig 2).

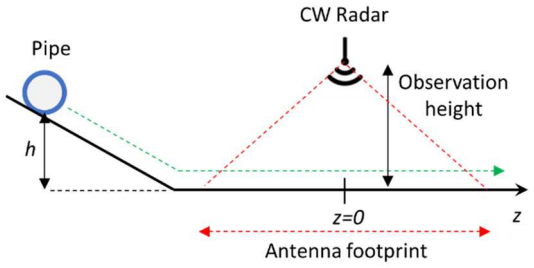


Fig. 2. Schematic diagram showing the initial and final positions of the circular pipe as it rolls through the radar beam. The radar was positioned at the centre of the horizontal plane at z = 0 m and z = -1.05 m from the initial position.

The velocity of the pipe v was calculated from the potential energy at the initial height h, converted to the velocity of the equated kinetic energy (both linear and angular):

$$v \cong \sqrt{gh} \quad (1)$$

where g is the gravitational acceleration. Note that the velocity is independent of the mass, diameter, and path length. This relationship is shown in Fig. 3. The initial heights ranged from 0 (no movement) to 300 mm with a velocity of up to 2.0 m/s. It is assumed that the rolling friction is minimal so that the velocity through the main beam is relatively constant for each starting height. For this reason, one can expect that the Doppler profiles will be symmetrical if the radiation pattern of the radar is symmetrical.

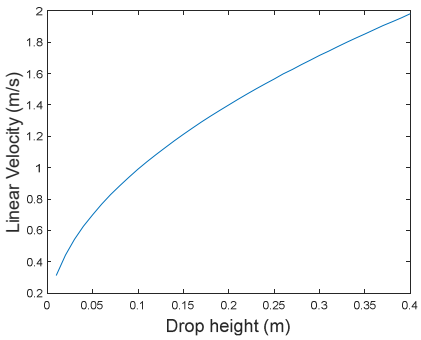
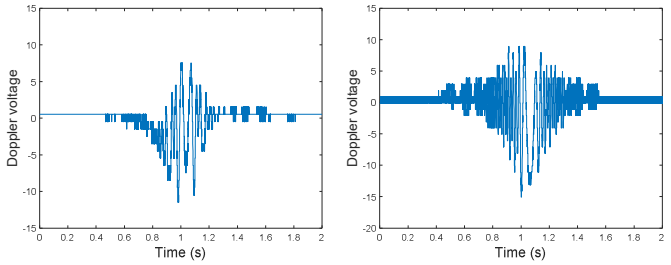


Fig. 3. The relationship between the linear velocity and drop height.

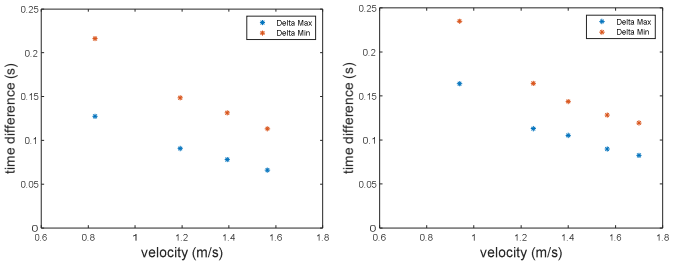
Figs. 4a and 4b show the radar voltage output as a function of time for the H-plane and the E-plane transits for an approximate pipe horizontal velocity of 1.6 m/s and 1.7 m/s respectively. Both H- and E-plane plots are approximately symmetrical on either side of the central minimum value.



(a) (b)

Fig. 4. (a) H-plane Doppler voltage as a function of time (v = 1.6 m/s). (b) E-plane Doppler voltage as a function of time (v = 1.7 m/s).

The time between the two maximum features symmetrically placed around the central minimum (H-plane) were measured as a function of the velocity v. Fig.5a shows the result in addition to the time between the first two minimum values adjacent to the central minimum. A similar relationship is evident for the E-plane case (Fig. 5b), however, the linearity is less evident. In both cases there is an apparent symmetry about the central minimum position when the pipe is directly beneath the radar nadir.



(a) (b)

Fig. 5. Time differences around the central minimum for the adjacent maximum and minimum values, respectively. (a) H-plane. (b) E-plane.

IV. Conclusions

The velocity of this large object can be determined providing the radar height above the plane remains constant. For this reason, velocity measurements using the HB100 radar can be accurately achieved for very low cost, very small infrastructure and low power. For example, the recording of ripple velocity (much smaller scatterers) in semi-filled, storm water pipes from this radar positioned at the highest point inside a pipe can provide water surface velocity measurements. This is the ultimate goal of this preliminary experiment.

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